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Axion datasheet (preliminary)

This is a preliminary datasheet and is subject to change.

Overview

The Axion is a global LTE CAT 1 bis module that supports cellular, Wi-Fi, and Bluetooth LE (BLE). It is based on the Realtek RTL8722DM (same as the M-SoM) and contains built-in battery charging circuitry in a convenient Feather form-factor that can be be used in prototyping breadboards or in production designs.

- Feather form-factor, like the Boron or Photon 2
- Can use cellular or Wi-Fi (2.4 GHz or 5 GHz) for the cloud connection
- Realtek RTL8722DM MCU (BLE and Wi-Fi)
- Quectel EG800Q-GL LTE CAT 1 bis (global) cellular modem

WHAT IS LTE CAT 1 BIS

LTE CAT 1 bis provides cellular connectivity using a compact and lower-cost cellular module and a single antenna.

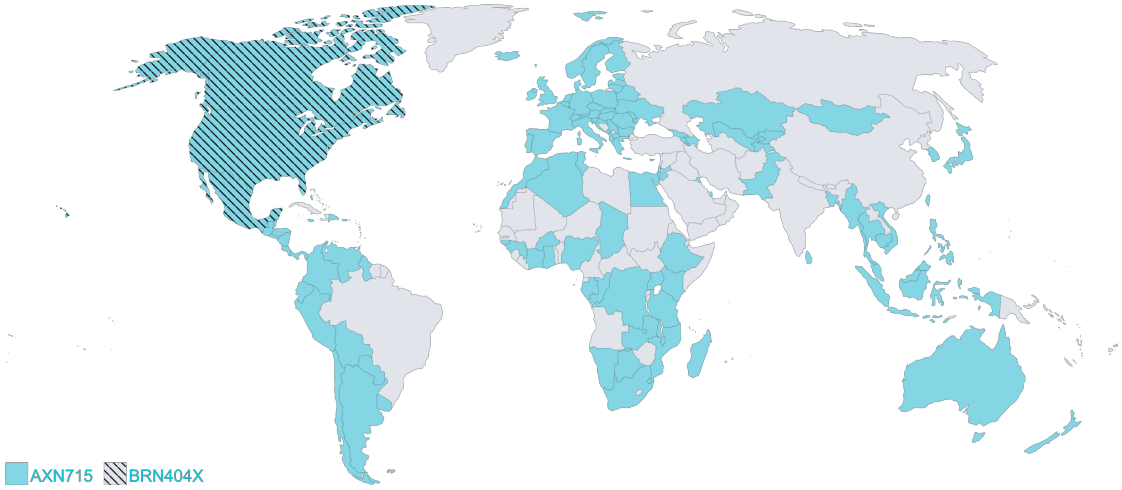
LTE CAT 1 (regular) modules require two antennas for receive antenna diversity, and generally require higher power, especially for 2G or 3G fallback.

LTE Cat M1 is also a compact, low-cost, single antenna standard, however it requires additional equipment at the cellular tower side and is not universally implemented worldwide.

LTE CAT 1 and LTE CAT 1 bis use the same 4G LTE towers used by 4G cell phones. Since all 5G cell phones also can connect to 4G, there is widespread support and no carriers have announced plans to sunset 4G.

COVERAGE COMPARISON

Since the Axion AXN715e supports LTE Cat 1 bis, it can be used world-wide with a single SKU. The Boron BRN404X is recommended only for use in the United States, Canada, and Mexico. The Axion also supports both Wi-Fi and cellular.



DEVICE COMPARISON

	Axion	Boron 404X	Photon 2	B504e	M404	M524
Form factor	Feather	Feather	Feather	M.2 SoM	M.2 SoM	M.2 SoM
Lifecycle	In development	Deprecated	GA	GA	GA	GA

Cellular modem	Quectel EG800Q-GL	u-blox SARA-R510S	None	Quectel EG91-NAX	Quectel BG95-M5	Quectel EG91-EX
Cellular region	Global	NorAm	N/A	Americas	NorAm	EMEAA
Cellular technology	LTE CAT 1 bis	LTE Cat M1	None	LTE Cat 1 with 3G fallback	LTE Cat M1	LTE Cat 1 with 2G/3G fallback
Wi-Fi supported	Yes (2.4 GHz and 5 GHz)	No	Yes (2.4 GHz and 5 GHz)	No	Yes (2.4 GHz and 5 GHz)	Yes (2.4 GHz and 5 GHz)
Generation	Gen 4	Gen 3	Gen 4	Gen 3	Gen 4	Gen 4
MCU	RTL8722DM	nRF52840	RTL8722DM	nRF52840	RTL8722DM	RTL8722DM
MCU processor speed	200 MHz Cortex-M33	64 MHz Cortex-M4F	200 MHz Cortex-M33	64 MHz Cortex-M4F	200 MHz Cortex-M33	200 MHz Cortex-M33
User RAM available	3072 KB	~80 KB	3072 KB	~80 KB	3072 KB	3072 KB
Maximum user application size	2048 KB	256 KB	2048 KB	256 KB	2048 KB	2048 KB
Li-Po battery connector	3-pin	2-pin	2-pin	No	No	No
RGB status LED	Yes	Yes	Yes	No[1]	No[1]	No[1]
MODE and RESET buttons	Yes	Yes	Yes	No[1]	No[1]	No[1]
USB connector	USB-C	Micro-B	Micro-B	No[1]	No[1]	No[1]
BLE support	Yes	Yes	Yes	Yes	Yes	Yes
NFC tag support	No	Yes	No	Yes	No	No
GNSS support	No	No	No	Yes	Yes	Yes
SIM	e-sim	MFF2 or 4FF	None	e-sim	MFF2	MFF2
Cellular antenna	PARANTCW1EA	PARANTC41EA, ANT-FLXU	None	PARANTCW1EA	PARANTCW1EA	PARANTCW1EA
Wi-Fi antenna	PARANTWM1EA	None	PARANTWM1EA[2]	None	PARANTWM1EA	PARANTWM1EA
BLE antenna	PARANTWM1EA	ANT-FLXV2	PARANTWM1EA[2]	ANT-FLXV2	PARANTWM1EA	PARANTWM1EA
On-board antennas	None	BLE PCB antenna	Wi-Fi/BLE PCB antenna	None	None	None
Platform	N/A	Boron (13)	Photon 2 / P2 (32)	B5-SoM (25)	M-SoM (35)	M-SoM (35)

- [1] not included on the M.2 SoM module, typically included on your base board.
- [2] Wi-Fi and BLE share a single external antenna.

Cellular Region:

- NorAm: United States, Canada, and Mexico.
- Americas: North, Central, and South America (not all countries supported).
- EMEAA: Europe, Middle East, Africa, and Asia (not all countries supported).

- See Country comparison below for additional information.

SIM:

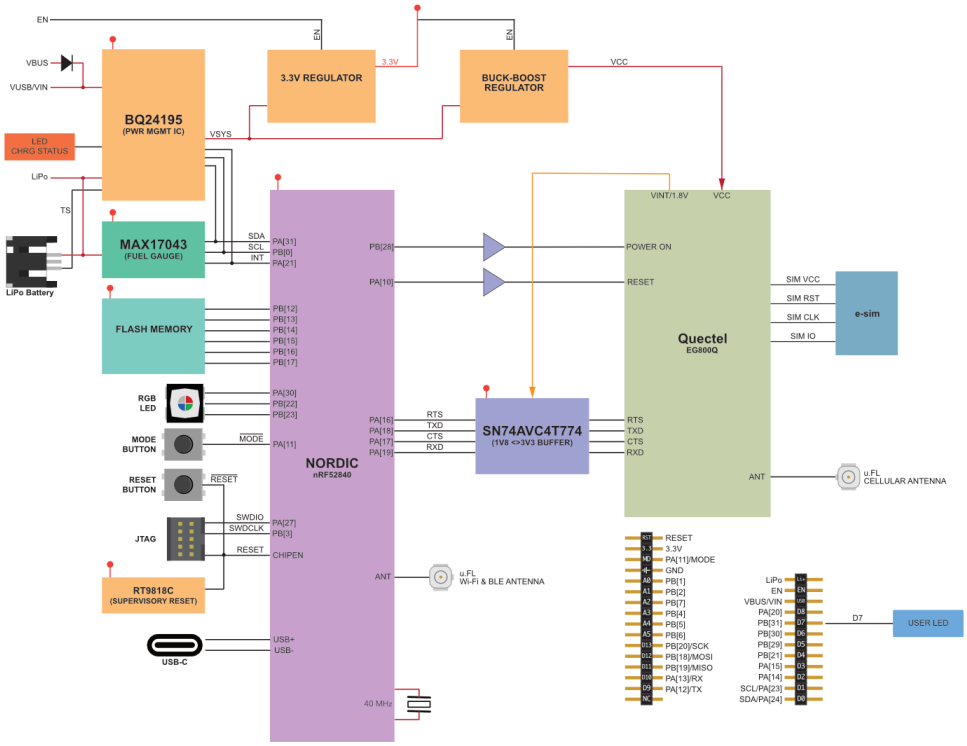
- e-sim: Electronic SIM included. Cannot be programmed with arbitrary 3rd-party profiles on these devices.
- MFF2: SMD SIM included, cannot be changed.
- 4FF: Plastic nano SIM card, can be used with 3rd-party SIM cards

MCU

The Realtek RTL8722DM is in the same family as the P2 and Photon 2 modules (RTL8721DM), but has additional GPIO.

- 802.11a/b/g/n Wi-Fi, 2.4 GHz and 5 GHz
  - U.FL connector for external antenna
- BLE 5 using same antenna as Wi-Fi
- Realtek RTL8722DM MCU
  - ARM Cortex M33 CPU, 200 MHz
- 2048 KB (2 MB) user application maximum size
- 3072 KB (3 MB) of RAM available to user applications
- 2 MB flash file system for user data storage
- FCC (United States), ISED (Canada), and CE (European Union) certified

BLOCK DIAGRAM



MIGRATION GUIDES

If you are migrating from a different Particle device to Axion the following migration guides may be helpful:

- Axion from Photon 2
- Axion from Boron

POWER

USB

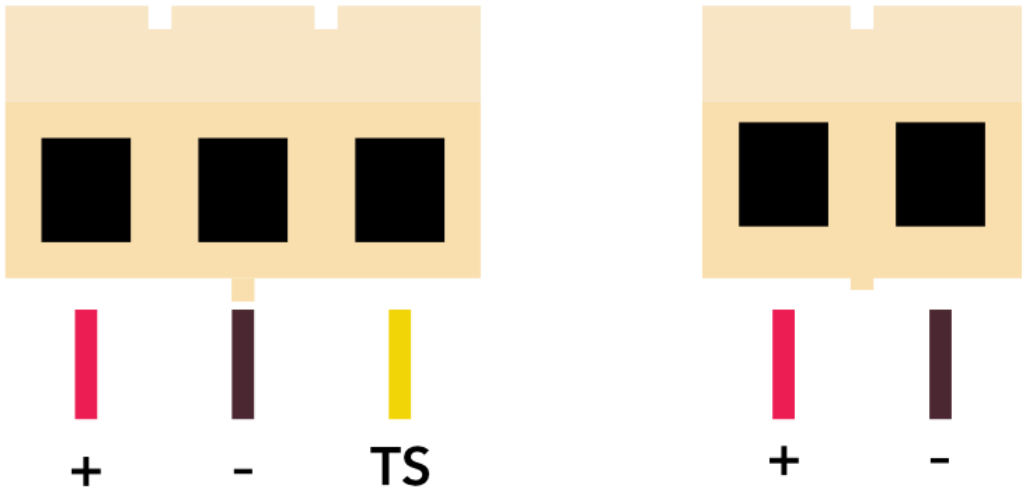
The USB port is the easiest way to power Axion. Please make sure that the USB port is able to provide at least 500mA. The connector is USB-C, which is different than the Photon 2 and Boron, which have a Micro B USB connector.

VUSB Pin

The pin is internally connected to the VBUS of the USB port. The nominal output should be around 4.5 to 5 VDC when the device is plugged into the USB port and 0 when not connected to a USB source. You can use this pin to power peripherals that operate at such voltages. Do not exceed the current rating of the USB port, which is nominally rated to 500mA.

LiPo

Axion includes a 3-pin JST-PH connector (left), not the 2-pin JST-PH connector on some other Particle devices (right). The TS pin is expected to be connected to a 10K NTC thermistor in the battery pack.



Looking at the exposed end of the connector attached to the battery

If purchasing a LiPo battery from a 3rd-party supplier, beware as the polarity of the JST-PH connector is not standardized and may be reversed. Permanent damage to can occur if powered by reverse polarity on the JST connector. See the battery guide for additional information.

Li+ pin

This pin is internally connected to the positive terminal of the LiPo connector. You can connect a single cell LiPo/Lithium Ion or a DC supply source to this pin for powering Axion. Remember that the input voltage range on this pin is 3.6 to 4.2 VDC.

3V3 PIN

This pin is the output of the on board 3.3V step-down switching regulator. The regulator is rated at 500mA max. When using this pin to power other devices or peripherals remember to budget in the current requirement of Axion first.

EN pin

The **EN** pin is not a power pin, per se, but it controls the 3V3 power. The EN pin is pulled high by a 100K resistor to the higher of VUSB, the USB connector, or Li+. Because the pull-up can result in voltages near 5V you should never directly connect EN to a 3.3V GPIO pin. Instead, you should only pull EN to GND, such as by using an N-channel MOSFET or other open-collector transistor.

The EN pin can force the device into a deep power-down state where it uses very little power. It can also used to assure that the device is completely reset, similar to unplugging it, with one caveat:

If using the EN pin to deeply reset the device, you must be careful not to allow leakage current back into the MCU by GPIO or by pull-ups to 3V3. If you only power external devices by 3V3 you won't run into this, as 3V3 is de-powered when EN is low.

However, if you have circuitry that is powered by a separate, external power supply, you must be careful. An externally powered circuit that drives a GPIO high when EN is low can provide enough current to keep the MCU from powering down and resetting. Likewise, a pull-up to an external power supply can do the same thing. Be sure that in no circumstances can power be supplied to the MCU when 3V3 is de-powered.

Antennas

- Axion includes two U.FL connectors for external antennas:
  - Cellular
  - Wi-Fi (2.4 GHz and 5 GHz) and BLE
- Wi-Fi operation in the 5150-5250 MHz band is only for indoor use to reduce the potential for harmful interference to co-channel mobile satellite systems.

Approved Antennas

CERTIFIED CELLULAR ANTENNAS

Axion is certified with the following cellular antenna:

Antenna	SKU	Details	Links
Wide band LTE cell antenna [x1]	PARANTCW1EA	B504e and M-SoM	Datasheet
Wide band LTE cell antenna [x50]	PARANTCW1TY	B504e and M-SoM	Datasheet

Single quantity units and developer kits include a PARANTCW1EA antenna. Tray quantities do not include antennas.

Dimension	Value	Unit
Length	116.0	mm
Width	27.0	mm
Thickness	0.2	mm
Cable Length	189.5	mm

Parameter	617-960	1156 - 1496	1520 - 1660.5	1710 - 2700	3330 - 4200	4200 - 5925
V.S.W.R.	≤ 3.5	≤ 3.0	≤ 2.0	≤ 2.0	≤ 1.5	≤ 2.0
Peak Gain	2.8 dBi	2.4 dBi	3.9 dBi	5.3 dBi	5.6 dBi	7.9 dBi
Eff % (max/avg)	71/62	55/53	68/62	85/70	76/71	69/62

CERTIFIED WI-FI/BLE ANTENNAS

Axion is certified for use with the same antennas as the P2/Photon 2. The same antenna is shared for Wi-Fi and BLE. Unlike the P2/Photon 2, the external antenna is required for Wi-Fi and BLE and Axion does not include a built-in trace antenna on the module.

Antenna	SKU	Links
Particle P2/Photon2 Wi-Fi Antenna 2.4/5GHz, [x1]	PARANTWM1EA	Datasheet | Retail Store
Particle P2/Photon2 Wi-Fi Antenna 2.4/5GHz, [x50]	PARANTWM1TY	Datasheet

Single quantity eveloper kits include a PARANTWM1EA antenna. Tray quantities do not include antennas.

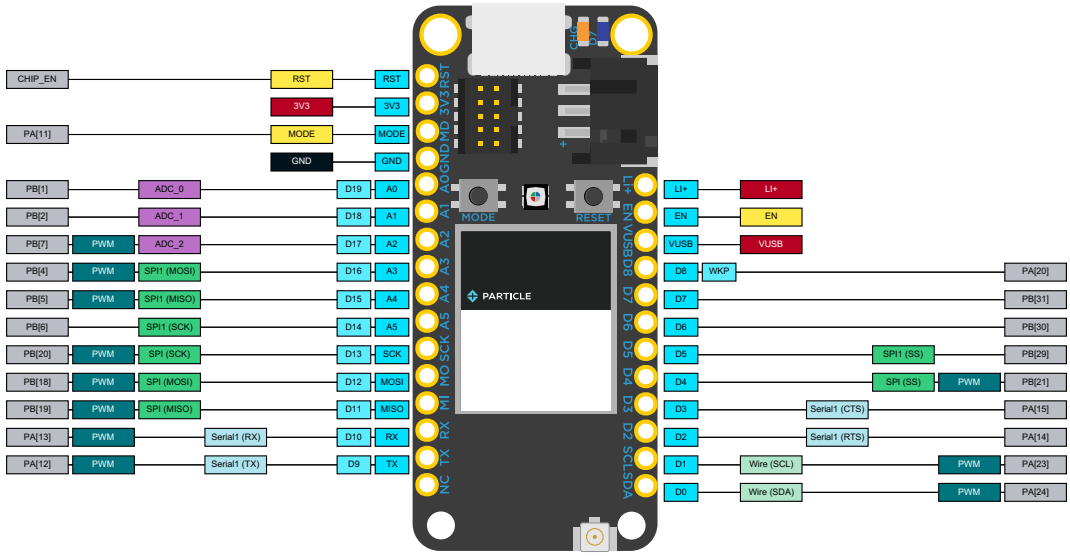
Do not use the Argon Wi-Fi/BLE antenna (ANT-FLXV2) on the P2, Photon 2, or M-SoM. The Argon antenna does not work with 5 GHz and this will result in poor Wi-Fi performance on the P2, Photon 2, and M-SoM.

GENERAL ANTENNA GUIDANCE

- The antenna placement needs to follow some basic rules, as any antenna is sensitive to its environment. Mount the antenna at least 10mm from metal components or surfaces, ideally 20mm for best radiation efficiency, and try to maintain a minimum of three directions free from obstructions to be able to operate effectively.
- Needs tuning with actual product enclosure and all components.

Pins and button definitions

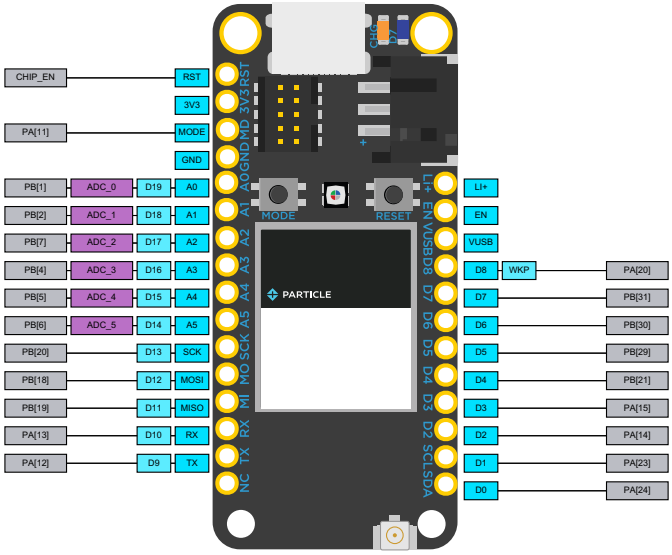
PINOUT DIAGRAM



GPIO AND PORT LISTING

Pin Name				PWM	MCU
A0 / D19	ADC_0				PB[1]
A1 / D18	ADC_1				PB[2]
A2 / D17	ADC_2			✓	PB[7]
A3 / D16	ADC_3	SPI1 (MOSI)		✓	PB[4]
A4 / D15	ADC_4	SPI1 (MISO)		✓	PB[5]
A5 / D14	ADC_5	SPI1 (SCK)			PB[6]
D0	Wire (SDA)			✓	PA[24]
D1	Wire (SCL)			✓	PA[23]
D2			Serial1 (RTS)		PA[14]
D3			Serial1 (CTS)		PA[15]
D4		SPI (SS)		✓	PB[21]
D5		SPI1 (SS)			PB[29]
D6					PB[30]
D7					PB[31]
D8 / WKP					PA[20]
MISO / D11		SPI (MISO)		✓	PB[19]
MOSI / D12		SPI (MOSI)		✓	PB[18]
RX / D10			Serial1 (RX)	✓	PA[13]
SCK / D13		SPI (SCK)		✓	PB[20]
TX / D9			Serial1 (TX)	✓	PA[12]

ADC (ANALOG TO DIGITAL CONVERTER)

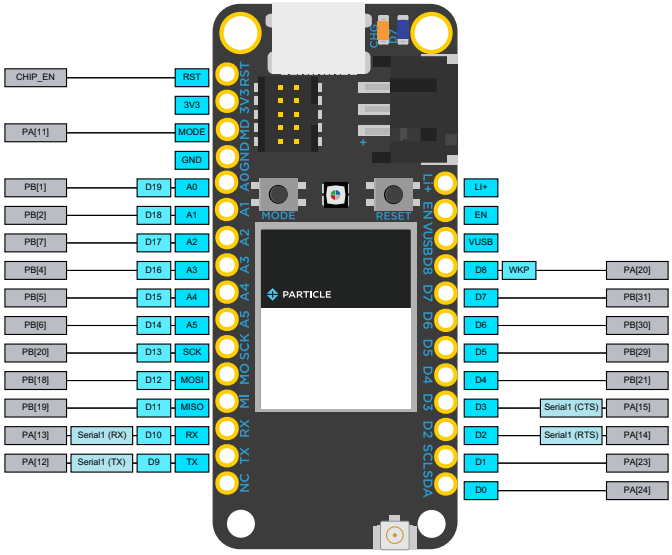


Axion supports 6 ADC inputs.

Pin Name	Description	Interface	MCU
A0 / D19	A0 Analog in, PDM CLK, GPIO	ADC_0	PB[1]
A1 / D18	A1 Analog in, PDM DAT, GPIO	ADC_1	PB[2]
A2 / D17	A2 Analog in, GPIO, PWM	ADC_2	PB[7]
A3 / D16	A3 Analog in, SPI1 MOSI, GPIO, PWM	ADC_3	PB[4]
A4 / D15	A4 Analog in, SPI1 MISO, GPIO, PWM	ADC_4	PB[5]
A5 / D14	A5 Analog in, SPI1 SCK, GPIO	ADC_5	PB[6]

- ADC inputs are single-ended and limited to 0 to 3.3V
- Resolution is 12 bits

UART SERIAL



Axion supports one UART serial interface.

Pin Name	Description	Interface	MCU
D2	D2 GPIO, Serial1 RTS flow control (optional)	Serial1 (RTS)	PA[14]
D3	D3 GPIO, Serial1 CTS flow control (optional)	Serial1 (CTS)	PA[15]
RX / D10	Serial RX, PWM, GPIO	Serial1 (RX)	PA[13]

TX / D9	Serial TX, PWM, GPIO, I2S MCLK	Serial1 (TX)	PA[12]

- The UART pins are 3.3V and must not be connected directly to a RS-232C port or to a 5V TTL serial port
- Hardware flow control is optional; if not used then the RTS and CTS pins can be used as regular GPIO

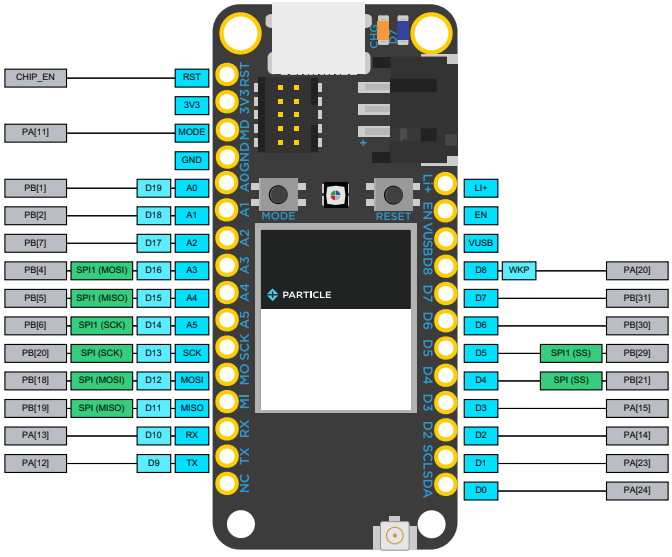
UART serial vs. Photon 2

The primary UART serial (`Serial1`) is on the TX and RX pins on both Axion and Photon 2. On Axion only, `Serial1` optionally supports hardware flow control. On the Photon 2 only, there are additional UART serial ports `Serial2` and `Serial3`.

Photon 2 Pin Name	Photon 2 Serial	Axion Pin Name	Axion Serial
SCK / D17	Serial3 (RTS)	SCK / D13	
MOSI / D15	Serial3 (TX)	MOSI / D12	
MISO / D16	Serial3 (RX)	MISO / D11	
RX / D9	Serial1 (RX)	RX / D10	Serial1 (RX)
TX / D8	Serial1 (TX)	TX / D9	Serial1 (TX)
D2	Serial2 (RTS)	D2	Serial1 (RTS)
D3	Serial2 (CTS)	D3	Serial1 (CTS)
D4	Serial2 (TX)	D4	
D5	Serial2 (RX)	D5	
D10 / WKP	Serial3 (CTS)	D8 / WKP	

SPI

Axion supports two SPI (serial peripheral interconnect) ports.



Pin Name	Description	Interface	MCU
A3 / D16	A3 Analog in, SPI1 MOSI, GPIO, PWM	SPI1 (MOSI)	PB[4]
A4 / D15	A4 Analog in, SPI1 MISO, GPIO, PWM	SPI1 (MISO)	PB[5]
A5 / D14	A5 Analog in, SPI1 SCK, GPIO	SPI1 (SCK)	PB[6]
D4	D4 GPIO, SPI SS, PWM, I2S WS	SPI (SS)	PB[21]
D5	D5 GPIO, SPI1 SS	SPI1 (SS)	PB[29]
MISO / D11	SPI MISO, D11 GPIO, PWM, I2S TX	SPI (MISO)	PB[19]

MOSI / D12	SPI MOSI, D12 GPIO, PWM	SPI (MOSI)	PB[18]
SCK / D13	D13 GPIO, SPI SCK, PWM, I2S CLK	SPI (SCK)	PB[20]

- The SPI port is 3.3V and must not be connected directly to devices that drive MISO at 5V
- If not using a SPI port, its pins can be used as GPIO
- Any pins can be used as the SPI chip select
- Multiple devices can generally share a single SPI port
- The primary SPI port has a maximum speed of 50 MHz and supports both SPI master and SPI slave modes
- The secondary SPI port (`SPI1`) has a maximum speed of 25 MHz and only supports SPI master mode

SPI vs. Boron

Both Axion and the Boron support two SPI interfaces, however `SPI1` (secondary SPI port) is on different pins.

Boron Pin Name	Boron SPI	Axion Pin Name	Axion SPI
A3 / D16		A3 / D16	SPI1 (MOSI)
A4 / D15		A4 / D15	SPI1 (MISO)
A5 / D14	SPI (SS)	A5 / D14	SPI1 (SCK)
SCK / D13	SPI (SCK)	SCK / D13	SPI (SCK)
MOSI / D12	SPI (MOSI)	MOSI / D12	SPI (MOSI)
MISO / D11	SPI (MISO)	MISO / D11	SPI (MISO)
D2	SPI1 (SCK)	D2	
D3	SPI1 (MOSI)	D3	
D4	SPI1 (MISO)	D4	SPI (SS)
D5		D5	SPI1 (SS)

SPI vs. Photon 2

Both Axion and the Photon 2 support two SPI interfaces, however there are important differences.

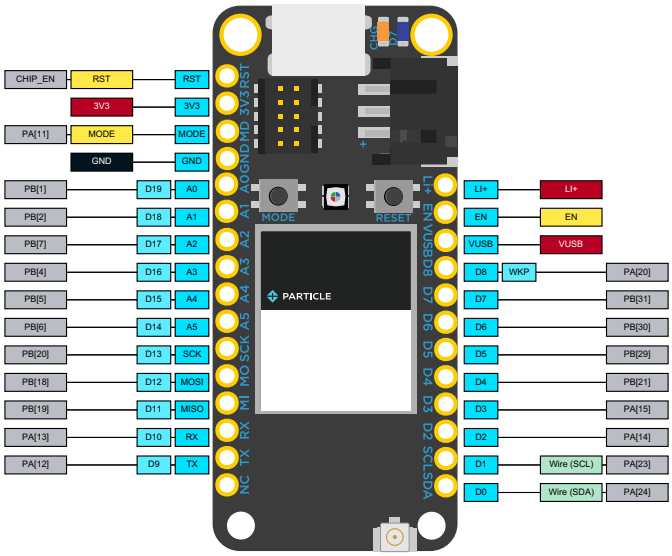
	Photon 2	Axion
`SPI`	25 MHz	50 MHz
SPI Master	✓	✓
SPI Slave		✓
Hardware peripheral	RTL872x SPI1	RTL872x SPI0
`SPI1`	50 MHz	25 MHz
SPI Master	✓	✓
SPI Slave	✓	
Pins	D2 - D5	A2 - A5
Hardware peripheral	RTL872x SPI0	RTL872x SPI1

Additionally, `SPI1` is on different pins.

Photon 2 Pin Name	Photon 2 SPI	Axion Pin Name	Axion SPI
A5 / D14		A3 / D16	SPI1 (MOSI)
S4 / D19		A4 / D15	SPI1 (MISO)
S3 / D18	SPI (SS)	A5 / D14	SPI1 (SCK)

SCK / D17	SPI (SCK)	SCK / D13	SPI (SCK)
MOSI / D15	SPI (MOSI)	MOSI / D12	SPI (MOSI)
MISO / D16	SPI (MISO)	MISO / D11	SPI (MISO)
D2	SPI1 (MOSI)	D2	
D3	SPI1 (MISO)	D3	
D4	SPI1 (SCK)	D4	SPI (SS)
D5	SPI1 (SS)	D5	SPI1 (SS)

I2C

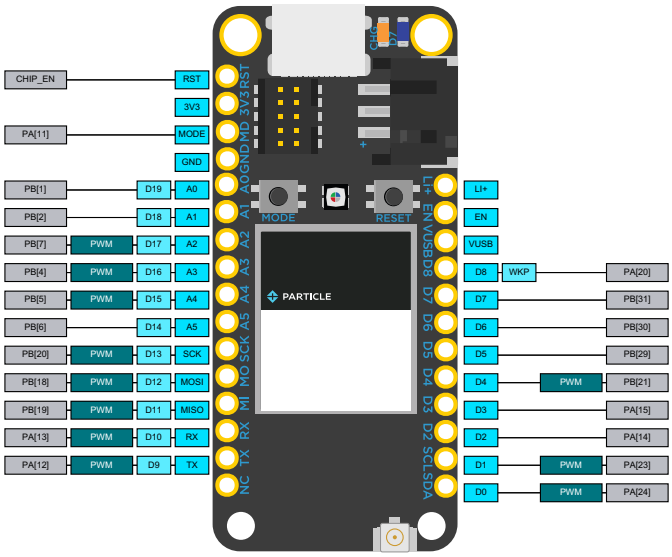


Axion supports one I2C (two-wire serial interface) port.

Pin Name	Description	Interface	MCU
D0	I2C SDA, GPIO, PWM	Wire (SDA)	PA[24]
D1	I2C SCL, GPIO, PWM	Wire (SCL)	PA[23]

- The I2C port is 3.3V and must not be connected directly to a 5V I2C bus
- Maximum bus speed is 400 kHz
- If not using I2C, pins D0 and D1 can be used as GPIO or analog input.
- The PMIC and Fuel Gauge are on a separate, dedicated I2C interface and do not affect `Wire`.

PWM



Axion supports PWM (pulse-width modulation) on the following pins:

Pin Name	Description	Timer	MCU
A2 / D17	A2 Analog in, GPIO, PWM		PB[7]
A3 / D16	A3 Analog in, SPI1 MOSI, GPIO, PWM		PB[4]
A4 / D15	A4 Analog in, SPI1 MISO, GPIO, PWM		PB[5]
D0	I2C SDA, GPIO, PWM		PA[24]
D1	I2C SCL, GPIO, PWM		PA[23]
D4	D4 GPIO, SPI SS, PWM, I2S WS		PB[21]
MISO / D11	SPI MISO, D11 GPIO, PWM, I2S TX		PB[19]
MOSI / D12	SPI MOSI, D12 GPIO, PWM		PB[18]
RX / D10	Serial RX, PWM, GPIO		PA[13]
SCK / D13	D13 GPIO, SPI SCK, PWM, I2S CLK		PB[20]
TX / D9	Serial TX, PWM, GPIO, I2S MCLK		PA[12]

On the RTL872x devices, all PWM pins and the RGB LED share the same PWM timer and must share the same PWM frequency, though each pin can have a separate duty cycle.

PWM vs. Boron

The pins that support PWM are different on the Boron (and Argon) and Axion.

Boron Pin Name	Boron PWM	Axion Pin Name	Axion PWM
A0 / D19	✓	A0 / D19	
A1 / D18	✓	A1 / D18	
A2 / D17	✓	A2 / D17	✓
A3 / D16	✓	A3 / D16	✓
A4 / D15	✓	A4 / D15	✓
A5 / D14	✓	A5 / D14	
SCK / D13		SCK / D13	✓
MOSI / D12		MOSI / D12	✓
MISO / D11		MISO / D11	✓
RX / D10		RX / D10	✓
TX / D09		TX / D9	✓
D0		D0	✓

D1		D1	✓
D2	✓	D2	
D3	✓	D3	
D4	✓	D4	✓
D5	✓	D5	
D6	✓	D6	
D7	✓	D7	
D8 / WKP	✓	D8 / WKP	

PWM vs. Photon 2

The pins that support PWM are different on the Photon 2 and Axion.

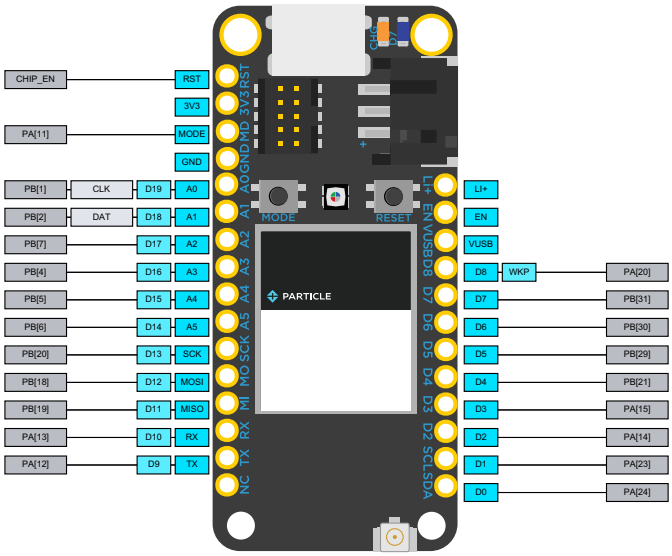
Photon 2 Pin Name	Photon 2 PWM	Axion Pin Name	Axion PWM
A2 / D13	✓	A2 / D17	✓
A5 / D14	✓	A3 / D16	✓
S4 / D19		A4 / D15	✓
SCK / D17		SCK / D13	✓
MOSI / D15	✓	MOSI / D12	✓
MISO / D16	✓	MISO / D11	✓
RX / D9		RX / D10	✓
TX / D8		TX / D9	✓
D0 / A3		D0	✓
D1 / A4	✓	D1	✓
D4		D4	✓

CAN (CONTROLLER AREA NETWORK)

Axion does not support CAN.

- The Tracker SoM includes CAN via a MCP25625 CAN interface with integrated transceiver.
- Both the MCP2515 and MCP25625 work with the library used on the Tracker and can be used to add CAN to Axion.

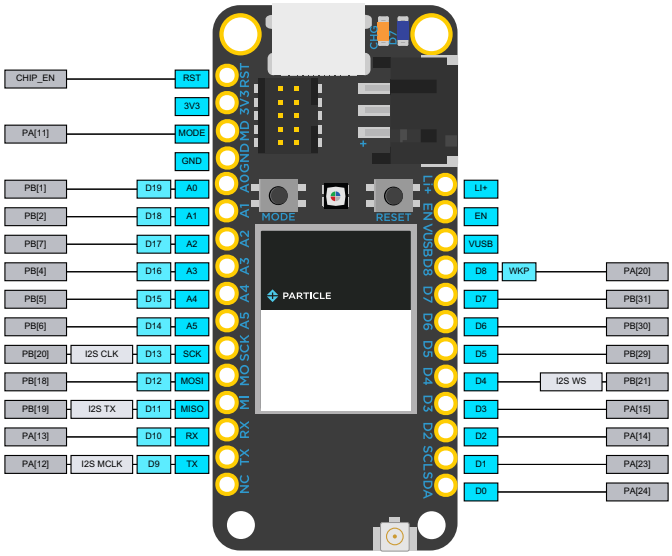
PDM



Pulse density modulation digital microphones can be used with the Microphone_PDM library and Axion, but only on specific pins.

Pin Name	Description	Interface	MCU
A0 / D19	A0 Analog in, PDM CLK, GPIO	CLK	PB[1]
A1 / D18	A1 Analog in, PDM DAT, GPIO	DAT	PB[2]

I2S (SOUND)



The Axion hardware supports I2S (sound), but there is no support for it in Device OS. It should be possible to implement in a third-party library from user firmware, but there is no publicly available library available at this time.

Pin Name	Description	Interface	MCU
D4	D4 GPIO, SPI SS, PWM, I2S WS	I2S WS	PB[21]
MISO / D11	SPI MISO, D11 GPIO, PWM, I2S TX	I2S TX	PB[19]
SCK / D13	D13 GPIO, SPI SCK, PWM, I2S CLK	I2S CLK	PB[20]
TX / D9	Serial TX, PWM, GPIO, I2S MCLK	I2S MCLK	PA[12]

BLE (BLUETOOTH LE)

BLE Central and BLE Peripheral modes are supported on Axion.

BLE long range (coded PHY) is not supported on Axion.

PMIC AND FUEL GAUGE

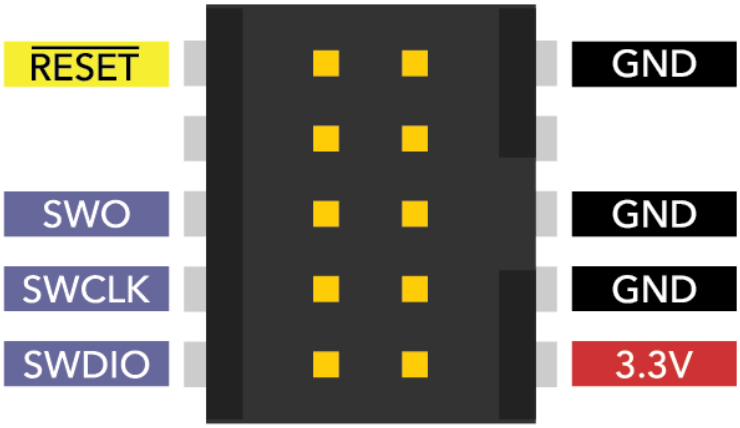
Axion contains the bq24195 PMIC and MAX17043 fuel gauge. They are on a dedicated I2C interface and will not interfere with the `Wire` interface on `D0` and `D1`.

Name	MCU	Description
PMIC_SDA	PB[0]	SDA for PMIC/Fuel Gauge
PMIC_SCL	PA[31]	SCL for PMIC/Fuel Gauge
LOW_BAT_UC	PA[21]	PMIC and Fuel Gauge interrupt

SWD

Axion has a dedicated 10 pin debug connector that exposes the SWD interface of the RTL872x. This interface can be used to debug your code or reprogram your bootloader, Device OS, or the user firmware using any standard SWD tools such as CMSIS/DAP debuggers.

	Axion	Photon 2	Boron
SWCLK	Dedicated	D6	Dedicated
SWDIO	Dedicated	D7	Dedicated
SWO	Not Used	Not Used	Dedicated



SWD vs. Photon 2

On the Photon 2, SWD is shared on D6 and D7. This is not the case on Axion, which uses a dedicated port for SWD, not shared with GPIO.

Photon 2 Pin Name	Photon 2 SWD	Axion Pin Name	Axion SWD
D6	SWCLK	D6	
D7	SWDIO	D7	

BOOT MODE PINS

Axion does not have RTL872x boot mode pins exposed on GPIO.

Boot mode pins vs. Photon 2

Because Axion uses dedicated pins for SWD, and uses the LP_UART for TX/RX instead of LOG_UART, the restrictions below for the Photon 2 do not apply to Axion.

Photon 2 Pin Name	Photon 2 Boot	Axion Pin Name	Axion Boot
TX / D8	Low at boot triggers ISP flash download	TX / D9	
D6	SWCLK. 40K pull-down at boot.	D6	
D7	SWDIO. 40K pull-up at boot. Low at boot triggers MCU test mode.	D7	

INTERRUPTS

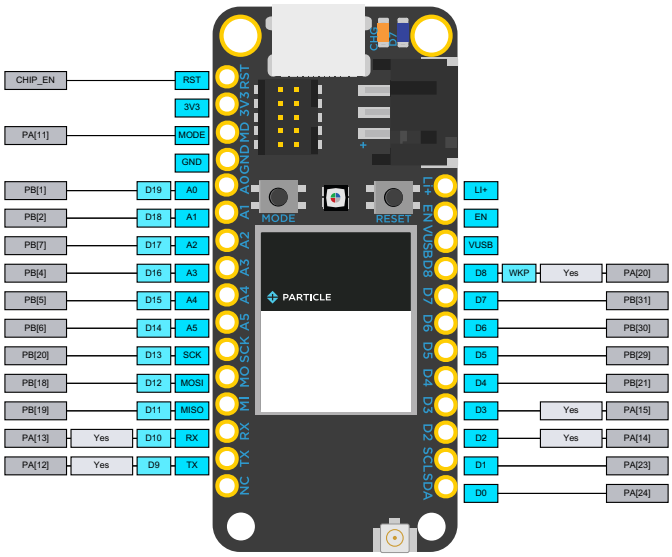
All pins can be used for interrupts on Axion.

There is a limit of 8 pin interrupts on the Boron; this limitation does not exist on Axion.

SLEEP

Axion can wake from `STOP` or `ULTRA_LOW_POWER` sleep mode on any GPIO, `RISING`, `FALLING`, or `CHANGE`.

Axion can only wake from `HIBERNATE` sleep mode on certain pins, `RISING`, `FALLING`, or `CHANGE` including pin D8, also known as `WKP`.



Pin	Pin Name	Description	Interface	MCU
14	RX / D10	Serial RX, PWM, GPIO	Pin can wake from HIBERNATE sleep	PA[13]
15	TX / D9	Serial TX, PWM, GPIO, I2S MCLK	Pin can wake from HIBERNATE sleep	PA[12]
18	D2	D2 GPIO, Serial1 RTS flow control (optional)	Pin can wake from HIBERNATE sleep	PA[14]
19	D3	D3 GPIO, Serial1 CTS flow control (optional)	Pin can wake from HIBERNATE sleep	PA[15]
24	D8 / WKP	GPIO, WKP	Pin can wake from HIBERNATE sleep	PA[20]

INTERNAL PULL-UP OR PULL-DOWN

Internal (MCU) pull-up and pull-down can be enabled using the `pinMode()` function and `INPUT_PULLUP` or `INPUT_PULLDOWN`. The pull-up or pull-down resistance varies by pin.

RETAINED MEMORY

Axion has limited support for retained memory, also referred to as Backup RAM or SRAM, in Device OS 5.3.1 and later.

Retained memory is preserved on RTL872x devices in the following cases:

Case	Saved
When entering sleep modes	5.3.1 and later
OTA firmware updates	5.3.1 and later
`System.backupRamSync()`	5.3.1 and later
`System.reset()`	Not saved
Reset button or reset pin	Not saved
Every 10 seconds	5.3.1 to 5.8.0 only

Calling `System.backupRamSync()` will manually save the contents of retained memory and will be restored after the device is reset. You should avoid saving the data extremely frequently as it is

slower than RAM and will cause flash wear and is relatively slow to execute. In Device OS 5.3.1 to 6.5.x it is saved to a dedicated flash page, and in Device OS 6.6.0 and later it is saved to the flash file system.

Prior to Device OS 5.3.1, retained memory is not supported on RTL872x devices. The flash file system can be used, or you can use an external chip such as an I2C or SPI FRAM.

Retained memory is 3068 bytes.

NFC TAG

Axion does not have NFC Tag support.

▶ Show pin details

1 RST

	Details
Pin Number	1
Pin Name	RST
Description	Hardware reset. Pull low to reset; can leave unconnected in normal operation.
MCU Pin	CHIP_EN

2 3V3

	Details
Pin Number	2
Pin Name	3V3
Description	Regulated 3.3V DC output

3 MODE

	Details
Pin Number	3
Pin Name	MODE
Description	MODE button, has internal pull-up
Supports attachInterrupt	Yes
MCU Pin	PA[11]

4 GND

	Details
Pin Number	4
Pin Name	GND
Description	Ground.

5 A0

	Details
Pin Number	5
Pin Name	A0
Pin Alternate Name	D19
Description	A0 Analog in, PDM CLK, GPIO
Supports digitalRead	Yes
Supports digitalWrite	Yes
Supports analogRead	Yes
Supports attachInterrupt	Yes
Internal pull resistance	2.1K
MCU Pin	PB[1]

6 A1

	Details
Pin Number	6
Pin Name	A1
Pin Alternate Name	D18
Description	A1 Analog in, PDM DAT, GPIO
Supports digitalRead	Yes
Supports digitalWrite	Yes
Supports analogRead	Yes
Supports attachInterrupt	Yes
Internal pull resistance	2.1K
MCU Pin	PB[2]

7 A2

	Details
Pin Number	7
Pin Name	A2
Pin Alternate Name	D17
Description	A2 Analog in, GPIO, PWM
Supports digitalRead	Yes
Supports digitalWrite	Yes
Supports analogRead	Yes
Supports analogWrite (PWM)	Yes
Supports tone	Yes
Supports attachInterrupt	Yes
Internal pull resistance	???
MCU Pin	PB[7]

8 A3

	Details
Pin Number	8
Pin Name	A3
Pin Alternate Name	D16
Description	A3 Analog in, SPI1 MOSI, GPIO, PWM
Supports digitalRead	Yes
Supports digitalWrite	Yes
Supports analogRead	Yes
Supports analogWrite (PWM)	Yes
Supports tone	Yes
SPI interface	MOSI. Use SPI1 object.
Supports attachInterrupt	Yes
Internal pull resistance	42K
MCU Pin	PB[4]

9 A4

	Details
Pin Number	9
Pin Name	A4
Pin Alternate Name	D15
Description	A4 Analog in, SPI1 MISO, GPIO, PWM
Supports digitalRead	Yes
Supports digitalWrite	Yes
Supports analogRead	Yes
Supports analogWrite (PWM)	Yes
Supports tone	Yes
SPI interface	MISO. Use SPI1 object.
Supports attachInterrupt	Yes
Internal pull resistance	???
MCU Pin	PB[5]

10 A5

	Details
Pin Number	10
Pin Name	A5
Pin Alternate Name	D14
Description	A5 Analog in, SPI1 SCK, GPIO
Supports digitalRead	Yes
Supports digitalWrite	Yes
Supports analogRead	Yes
SPI interface	SCK. Use SPI1 object.
Supports attachInterrupt	Yes
Internal pull resistance	22K
MCU Pin	PB[6]

11 SCK

	Details
Pin Number	11
Pin Name	SCK
Pin Alternate Name	D13
Description	D13 GPIO, SPI SCK, PWM, I2S CLK
Supports digitalRead	Yes
Supports digitalWrite	Yes
Supports analogWrite (PWM)	Yes
Supports tone	Yes
SPI interface	SCK. Use SPI object.
Supports attachInterrupt	Yes
I2S interface	I2S CLK
Internal pull resistance	???
MCU Pin	PB[20]

12 MOSI

	Details
Pin Number	12
Pin Name	MOSI
Pin Alternate Name	D12
Description	SPI MOSI, D12 GPIO, PWM
Supports digitalRead	Yes
Supports digitalWrite	Yes
Supports analogWrite (PWM)	Yes
Supports tone	Yes
SPI interface	MOSI. Use SPI object.
Supports attachInterrupt	Yes
Internal pull resistance	???
MCU Pin	PB[18]

13 MISO

	Details
Pin Number	13
Pin Name	MISO
Pin Alternate Name	D11
Description	SPI MISO, D11 GPIO, PWM, I2S TX
Supports digitalRead	Yes
Supports digitalWrite	Yes
Supports analogWrite (PWM)	Yes
Supports tone	Yes
SPI interface	MISO. Use SPI object.
Supports attachInterrupt	Yes
I2S interface	I2S TX
Internal pull resistance	???
MCU Pin	PB[19]

14 RX

	Details
Pin Number	14
Pin Name	RX
Pin Alternate Name	D10
Description	Serial RX, PWM, GPIO
Supports digitalRead	Yes
Supports digitalWrite	Yes
Supports analogWrite (PWM)	Yes
Supports tone	Yes
UART serial	RX. Use Serial1 object.
Supports attachInterrupt	Yes
Internal pull resistance	2.1K
MCU Pin	PA[13]

15 TX

	Details
Pin Number	15
Pin Name	TX
Pin Alternate Name	D9
Description	Serial TX, PWM, GPIO, I2S MCLK
Supports digitalRead	Yes
Supports digitalWrite	Yes
Supports analogWrite (PWM)	Yes
Supports tone	Yes
UART serial	TX. Use Serial1 object.
Supports attachInterrupt	Yes
I2S interface	I2S MCLK
Internal pull resistance	2.1K
MCU Pin	PA[12]

16 D0

	Details
Pin Number	16
Pin Name	D0
Description	I2C SDA, GPIO, PWM
Supports digitalRead	Yes
Supports digitalWrite	Yes
Supports analogWrite (PWM)	Yes
Supports tone	Yes
I2C interface	SDA. Use Wire object.
Supports attachInterrupt	Yes
Internal pull resistance	???
MCU Pin	PA[24]

17 D1

	Details
Pin Number	17
Pin Name	D1
Description	I2C SCL, GPIO, PWM
Supports digitalRead	Yes
Supports digitalWrite	Yes
Supports analogWrite (PWM)	Yes
Supports tone	Yes
I2C interface	SCL. Use Wire object.
Supports attachInterrupt	Yes
Internal pull resistance	???
MCU Pin	PA[23]

18 D2

	Details
Pin Number	18
Pin Name	D2
Description	D2 GPIO, Serial1 RTS flow control (optional)
Supports digitalRead	Yes
Supports digitalWrite	Yes
UART serial	RTS. Use Serial1 object.
Supports attachInterrupt	Yes
Internal pull resistance	???
MCU Pin	PA[14]

19 D3

	Details
Pin Number	19
Pin Name	D3
Description	D3 GPIO, Serial1 CTS flow control (optional)
Supports digitalRead	Yes
Supports digitalWrite	Yes
UART serial	CTS. Use Serial1 object.
Supports attachInterrupt	Yes
Internal pull resistance	???
MCU Pin	PA[15]

20 D4

	Details
Pin Number	20
Pin Name	D4
Description	D4 GPIO, SPI SS, PWM, I2S WS
Supports digitalRead	Yes
Supports digitalWrite	Yes
Supports analogWrite (PWM)	Yes
Supports tone	Yes
SPI interface	SS. Use SPI object.
Supports attachInterrupt	Yes
I2S interface	I2S WS
Internal pull resistance	???
MCU Pin	PB[21]

21 D5

	Details
Pin Number	21
Pin Name	D5
Description	D5 GPIO, SPI1 SS
Supports digitalRead	Yes
Supports digitalWrite	Yes

SPI interface	SS. Use SPI1 object.
Supports attachInterrupt	Yes
Internal pull resistance	???
MCU Pin	PB[29]

22 D6

	Details
Pin Number	22
Pin Name	D6
Description	GPIO
Supports digitalRead	Yes
Supports digitalWrite	Yes
Supports attachInterrupt	Yes
Internal pull resistance	???
MCU Pin	PB[30]

23 D7

	Details
Pin Number	23
Pin Name	D7
Description	GPIO
Supports digitalRead	Yes
Supports digitalWrite	Yes
Supports attachInterrupt	Yes
Internal pull resistance	13K
MCU Pin	PB[31]

24 D8

	Details
Pin Number	24
Pin Name	D8
Pin Alternate Name	WKP
Description	GPIO, WKP
Supports digitalRead	Yes
Supports digitalWrite	Yes
Supports attachInterrupt	Yes
Internal pull resistance	???
MCU Pin	PA[20]

25 VUSB

	Details
Pin Number	25
Pin Name	VUSB
Description	Power out (when powered by USB) 5 VDC, power in with limitations.
Input is 5V Tolerant	Yes

26 EN

	Details
Pin Number	26
Pin Name	EN
Description	Power supply enable. Connect to GND to power down. Has internal weak (100K) pull-up.

27 LI+

	Details
Pin Number	27
Pin Name	LI+
Description	Connected to JST PH LiPo battery connector. 3.7V in or out.

Technical specification

RECOMMENDED OPERATING CONDITIONS

Parameter	Symbol	Min	Typ	Max	Unit
Operating Temperature	T_{op}	-20		+70	°C
Humidity Range Non condensing, relative humidity				95	%

I/O CHARACTERISTICS

Parameter	Symbol	Min	Typ	Peak	Unit
Input-High Voltage	V_{IH}	2.0			V
Input-Low Voltage	V_{IL}			0.8	V
Output drive strength (normal drive)				4	mA
Output drive strength (high drive)				12	mA
Total output drive (all pins)				200	mA

POWER CONSUMPTION

To be provided at a later date

Mechanical specifications

To be provided at a later date

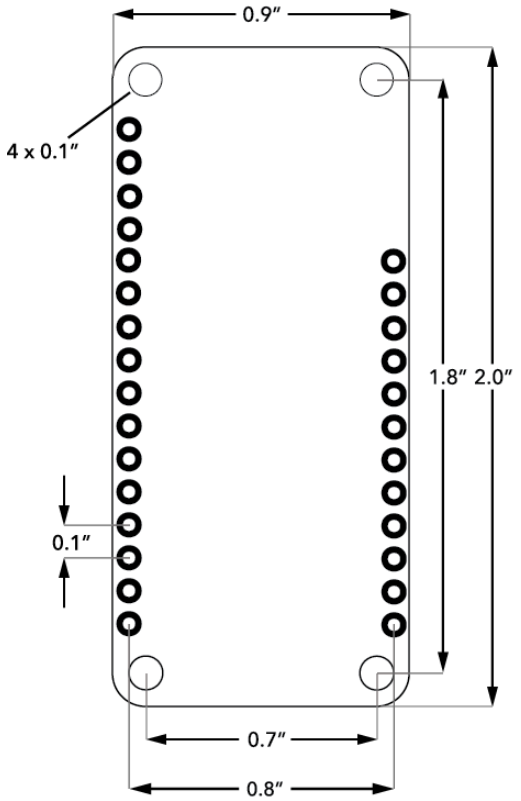
MATING CONNECTORS

Axion uses two single row 0.1" pitch male header pins. One of them is 16 pin while the other is 12 pin. It can be mounted with matching 0.1" pitch female headers with a typical height of 0.335" (8.5mm). When you search for parts like these it can be difficult to navigate the thousands of parts available online so here are a few good choices for Axion, the Photon 2, and Boron:

Description	MFG	MFG Part Number
16-pin 0.1" (2.54mm) Female Header	Sullins	PPTC161LFBN-RC
16-pin 0.1" (2.54mm) Female Header	TE	6-535541-4
12-pin 0.1" (2.54mm) Female Header	Sullins	PPTC121LFBN-RC
12-pin 0.1" (2.54mm) Female Header	TE	6-534237-0

RECOMMENDED PCB LAND PATTERN

Axion can be directly soldered onto the PCB or be mounted with the above mentioned female headers.



Assembly

WATER SOLUBLE FLUX

If you are attaching Axion to a custom base board, we recommend using a socket. As there are components on the bottom side of Axion there is no version available with castellated holes, solder pads, or similar techniques for direct surface mounting.

The pin headers on the bottom of Axion are not intended to be reflowed using paste-in-hole.

If you wish to surface mount module, you should directly use the P2 module on your base board instead of incorporating Axion.

If you decide to wave solder or hand-solder Axion directly to your base board, water soluble flux should not be used. There are components within the P2 module that are moisture-sensitive, and wash water can get trapped under the RF shields, causing damage.

Use no-clean flux instead if you must solder Axion module.

CONFORMAL COATINGS

Axion modules should not use a conformal coating to protect the module from water. Some components on the module cannot be coated and would need to be masked off during coating. This will make the coating process difficult to implement and test.

Furthermore, the buttons cannot be protected by using a coating. Using an enclosure that protects both your base board and Axion module as a single waterproof assembly is recommended instead.

Ordering information

SKU	Description	Region	Modem	EtherSIM	Lifecycle	Replacement
AXN715EHEA	Axion LTE CAT 1 bis (Global, EtherSIM+), [x1]	Global	EG800Q-GL	✓	In development	
AXN715EHTY	Axion LTE CAT 1 bis (Global, EtherSIM+), [x50]	Global	EG800Q-GL	✓	In development	

Qualification and approvals

To be provided at a later date

Product handling

MOISTURE SENSITIVITY LEVELS

The Moisture Sensitivity Level (MSL) relates to the packaging and handling precautions required. The P1 module is rated level 3. In general, this precaution applies for Photons without headers. When reflowing a P1 directly onto an application PCB, increased moisture levels prior to reflow can damage sensitive electronics on the P1. A bake process to reduce moisture may be required.

For more information regarding moisture sensitivity levels, labeling, storage and drying see the MSL standard see IPC/JEDEC J-STD-020 (can be downloaded from www.jedec.org).

ESD PRECAUTIONS

The P1 module contains highly sensitive electronic circuitry and is an Electrostatic Sensitive Device (ESD). Handling a P1 module without proper ESD protection may destroy or damage it permanently. Proper ESD handling and packaging procedures must be applied throughout the processing, handling and operation of any application that incorporates P1 modules. ESD precautions should be implemented on the application board where the P1 module is mounted. Failure to observe these precautions can result in severe damage to the P1 module!

Default settings

The P2 module comes pre-programmed with a bootloader and a user application called Tinker. This application works with an iOS and Android app also named Tinker that allows you to very easily toggle digital pins, take analog and digital readings and drive variable PWM outputs.

The bootloader allows you to easily update the user application via several different methods, USB, OTA, Serial Y-Modem, and also internally via the Factory Reset procedure. All of these methods have multiple tools associated with them as well.

You may use the Particle Web IDE to code, compile and flash a user application OTA (Over The Air). Particle Workbench is a full-featured desktop IDE for Windows, Mac, and Linux based on VSCode and supports both cloud-based and local gcc-arm compiles. The Particle CLI provides a command-line interface for cloud-based compiles and flashing code over USB.

Country compatibility

Country	Technologies	Carriers
Albania	4G	Eagle, Telekom, Vodafone
Algeria	4G	Mobilis, Ooredoo
Anguilla	4G	Flow
Antigua and Barbuda	4G	Flow
Argentina	4G	Claro, Movistar, Personal
Armenia	4G	Ucom
Aruba	4G	Setar
Australia	4G	Optus, Telstra, Vodafone
Austria	4G	3 (Drei), A1, T-Mobile
Azerbaijan	4G	Azercell, Bakcell, NAR Mobile
Bahamas	4G	Aliv, BTC Bahamas
Bahrain	4G	Zain
Bangladesh	4G	Bangalink, GrameenPhone
Barbados	4G	Flow
Belarus	4G	A1
Belgium	4G	Base, Orange, Proximus
Bolivia	4G	Viva
Botswana	4G	BeMobile
Brunei	4G	DST
Bulgaria	4G	A1, Telenor, Vivacom
Burkina Faso	4G	Orange
Cabo Verde	4G	CVMóvel, Unitel T+
Cambodia	4G	Metfone
Canada	4G	Bell Mobility, Rogers Wireless, Telus, Videotron
Cayman Islands	4G	Flow
Chad	4G	Airtel
Chile	4G	Claro, Entel, Movistar
Colombia	4G	Movistar, Tigo
Congo (Brazzaville)	4G	Airtel
Congo (Kinshasa)	4G	Airtel
Costa Rica	4G	Movistar
Côte d'Ivoire	4G	MTN
Croatia	4G	Hrvatski Telekom, Tele2
Cyprus	4G	Cytamobile-Vodafone, MTN, PrimeTel
Czechia	4G	O2, T-Mobile, Vodafone
Denmark	4G	3 (Tre), TDC, Telenor, Telia
Dominica	4G	Flow
Dominican Republic	4G	Altice Dominicana, Claro, Viva
Ecuador	4G	Claro, Movistar
Egypt	4G	Etisalat, Orange
El Salvador	4G	Claro, Telefonica
Estonia	4G	Elisa, Tele2, Telia

eSwatini	4G	MTN
Faroe Islands	4G	Faroese Telecom, Vodafone
Finland	4G	DNA, Elisa, Telia
France	4G	Bouygues, Free Mobile, Orange, SFR
Gabon	4G	Airtel
Georgia	4G	Beeline, Geocell
Germany	4G	O2, Telekom, Vodafone
Ghana	4G	MTN, Vodafone
Gibraltar	4G	Gibtel
Greece	4G	Cosmote, Vodafone, Wind
Guadeloupe	4G	Orange
Guatemala	4G	Claro, Movistar
Guernsey	4G	Sure
Guinea	4G	MTN
Guinea-Bissau	4G	MTN
Guyana	4G	Digicel
Honduras	4G	Claro, Tigo
Hong Kong	4G	CMHK, CSL, SmarTone
Hungary	4G	Magyar Telekom, Telenor, Vodafone
Iceland	4G	Nova, Siminn, Vodafone
Indonesia	4G	Indosat, Telkomsel, XL Axiata
Ireland	4G	3 (Tre), Meteor, O2, Vodafone
Isle of Man	4G	Manx Telecom, Sure
Israel	4G	Hot Mobile, Orange, Pelephone
Italy	4G	TIM, Vodafone, Wind
Jamaica	4G	Digicel, Flow
Japan	4G	KDDI, NTT DoCoMo, Softbank
Jersey	4G	Jersey Telecom, Sure
Jordan	4G	Zain
Kazakhstan	4G	Beeline, K-Cell
Kenya	4G	Airtel
Kuwait	4G	Viva, Zain
Kyrgyzstan	4G	Beeline
Latvia	4G	Bite, LMT, Tele2
Liechtenstein	4G	Mobilkom, Orange
Lithuania	4G	Bite, Omnitel, Tele2
Luxembourg	4G	Orange, POST, Tango
Madagascar	4G	Airtel
Malawi	4G	Airtel
Malaysia	4G	Celcom, DiGi, Maxis
Malta	4G	Go Mobile, Vodafone
Mexico	4G	AT&T, Telcel
Moldova	4G	Moldcell, Orange
Mongolia	4G	Mobicom, Unitel
Montenegro	4G	Mtel, T-Mobile, Telenor

Morocco	4G	Inwi, Medi Telecom
Mozambique	4G	Vodacom
Myanmar	4G	MPT, Telenor
Namibia	4G	Telecom Namibia
Netherlands	4G	KPN, T-Mobile, Vodafone
New Zealand	4G	2degrees, Spark, Vodafone
Nicaragua	4G	Movistar
Nigeria	4G	Airtel, MTN
North Macedonia	4G	Vip operator
Norway	4G	TDC, Telenor, Telia
Pakistan	4G	Jazz, Telenor, Ufone
Panama	4G	Movistar
Paraguay	4G	Claro, Personal, Tigo, Vox
Peru	4G	Claro, Entel, Movistar
Philippines	4G	Globe, Smart
Poland	4G	Orange, Play, Plus, T-Mobile
Portugal	4G	NOS, TMN, Vodafone
Puerto Rico	4G	Claro
Qatar	4G	Ooredoo, Vodafone
Romania	4G	Orange, Telekom Romania, Vodafone
Saint Kitts and Nevis	4G	Flow
Saint Lucia	4G	Flow
Saint Vincent and the Grenadines	4G	Flow
Serbia	4G	Telenor, VIP
Seychelles	4G	Airtel
Slovakia	4G	O2, Orange, Telekom
Slovenia	4G	A1, Mobitel
South Africa	4G	MTN, Vodacom
South Korea	4G	KT, LG U+, SK Telecom
Spain	4G	Orange, Telefonica, Vodafone, Yoigo
Sri Lanka	4G	Dialog, Mobitel
Sweden	4G	3 (Tre), Tele2, Telenor, Telia
Switzerland	4G	Salt, Sunrise
Taiwan	4G	Chunghwa, FarEasTone, T Star, Taiwan Mobile
Tajikistan	4G	Beeline, Tcell
Tanzania	4G	Airtel
Thailand	4G	AIS, DTAC, True Move
Trinidad and Tobago	4G	Digicel, TSTT
Tunisia	4G	Orange Tunisie, Tunisie Telecom
Turks and Caicos Islands	4G	Flow
Uganda	4G	Africell, Airtel, MTN
Ukraine	4G	Kyivstar, Life, MTS
United Kingdom	4G	3, EE, O2, Vodafone
United States	4G	Alaska Wireless, AT&T, T-Mobile (USA), Verizon[7]
Uruguay	4G	Antel, Claro, Movistar

Uzbekistan	4G	Beeline
Venezuela	4G	Movistar
Vietnam	4G	MobiFone, Viettel, Vinaphone
Virgin Islands (British)	4G	CCT, Flow
Virgin Islands (U.S.)	4G	T-Mobile (USA)
Zambia	4G	Airtel

COUNTRY COMPARISON

Country	Axion	BRN404X	B504e	M404	M524
Afghanistan					
Albania					
Algeria					
Anguilla					
Antigua and Barbuda					
Argentina					
Armenia					
Aruba					
Australia					
Austria					
Azerbaijan					
Bahamas					
Bahrain					
Bangladesh					
Barbados					
Belarus					
Belgium					
Belize					
Bolivia					
Bosnia and Herzegovina					
Botswana					
Brunei					
Bulgaria					
Burkina Faso					
Cabo Verde					
Cambodia					
Canada					
Cayman Islands					
Chad					
Chile					
Colombia					
Congo (Brazzaville)					
Congo (Kinshasa)					
Costa Rica					
Côte d'Ivoire					
Croatia					

Cyprus
Czechia
Denmark
Dominica
Dominican Republic
Ecuador
Egypt
El Salvador
Estonia
eSwatini
Ethiopia
Faroe Islands
Finland
France
French Guiana
Gabon
Georgia
Germany
Ghana
Gibraltar
Greece
Guadeloupe
Guatemala
Guernsey
Guinea
Guinea-Bissau
Guyana
Haiti
Honduras
Hong Kong
Hungary
Iceland
Indonesia
Ireland
Isle of Man
Israel
Italy
Jamaica
Japan
Jersey
Jordan
Kazakhstan
Kenya
Kuwait
Kyrgyzstan

Latvia
Liechtenstein
Lithuania
Luxembourg
Macao
Madagascar
Malawi
Malaysia
Malta
Mexico
Moldova
Mongolia
Montenegro
Morocco
Mozambique
Myanmar
Namibia
Netherlands
New Zealand
Nicaragua
Nigeria
North Macedonia
Norway
Pakistan
Palestine
Panama
Papua New Guinea
Paraguay
Peru
Philippines
Poland
Portugal
Puerto Rico
Qatar
Romania
Rwanda
Saint Kitts and Nevis
Saint Lucia
Saint Vincent and the Grenadines
Serbia
Seychelles
Sint Maarten
Slovakia
Slovenia
South Africa

South Korea
South Sudan
Spain
Sri Lanka
Suriname
Sweden
Switzerland
Taiwan
Tajikistan
Tanzania
Thailand
Trinidad and Tobago
Tunisia
Turks and Caicos Islands
Uganda
Ukraine
United Kingdom
United States
Uruguay
Uzbekistan
Venezuela
Vietnam
Virgin Islands (British)
Virgin Islands (U.S.)
Zambia